

Punjaya Wickramasinghe

Software Engineer · AI Safety

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Australia · Open to international relocation

Engineer with a co-authored journal paper, who has built and evaluated agentic LLM systems. Looking to work in AI safety, especially technical research supporting governance, and helping non-technical people understand AI risk.

AI SAFETY

Technical AI Safety Project
BlueDot Impact | Oct 2026 – Present

ControlArena · Inspect · Claude Code

- Evaluating backdoor monitors for AI control, comparing safety against cost across audit budgets
- Varying monitor model size and reasoning ability to find the most cost-effective monitoring setups

The Pivot's Archive
Substack · thepivotsarchive.substack.com | Sep 2026 – Present

- Writing to help non-technical people understand both the pros and cons of AI, using relatable, digestible analogies

WORK EXPERIENCE

Software Engineering Lead
Phoenix ICT Solutions | Jan 2025 – Mar 2026

LangChain · Ollama · FastAPI · Docker · VirtualBox · Flutter

- Co-led a 5-person team building Zara LM, an air-gapped LLM agent platform (RAG Q&A, cited report generation, analysis) to accelerate user workflows
- Built the eval pipeline to uplift performance and safety: groundedness 81% to 100%, context recall 48% to 77%, correctness 91% to 94%
- Gave the team and our CEO a shared source of truth, and the confidence to demo and deploy it to our first pilot users
- Maintained Australian Government security clearance

Software Engineering Fellow
Engineering for Change & Resham Sutra | May 2025 – Oct 2025

Java · Figma · Python · C++ · Whisper.cpp

- Shipped an offline voice assistant in GramSootra (Resham Sutra's textile marketplace Android app) for low-literacy artisans, compatible with low-end phones (40MB package)
- Human-centered design to overcome language and literacy barriers

Mechatronics Engineer
Geodrones Australia | Nov 2023 – Dec 2024

C++ · ROS · Gazebo · Docker · Altium · STM32 · I2C · PyTorch

- Finalist, Australian AI Awards 2024, and pitched the demo to industry VIPs at October Sky 2024
- Deployed a real-time AI change detection platform, fusing camera/LiDAR
- Developed secure real-time task swapping and adapted an open-source GPS-denied state estimator for our drones
- Wrote STM32 firmware to configure the battery charger in a battery management system, over I2C

PUBLICATIONS

Constructive synchronous observer design for inertial navigation with delayed GNSS measurements
European Journal of Control | 2024

P. van Goor, **P. Wickramasinghe**, M. Hampsey, R. Mahony · doi.org/10.1016/j.ejcon.2024.101047

Honours thesis (ANU, 2023), supervised by two of the paper's authors

Lie Theory · Python · ArduPilot · LaTeX

- Developed and implemented a nonlinear inertial navigation filter in Python with provable stability that compensates for GPS time delay and uses magnetometer data; tested in simulation and on a real VTOL flight log replayed in ArduPilot
- Thesis work informed the co-authored journal paper, which extends the theory with full stability proofs
- Early results were shared by my supervisor at the ArduPilot 2023 Conference

EDUCATION

Bachelor of Engineering (Research & Development)
Australian National University (ANU) | 2019 – 2023

- Mechatronics & Computer Science · First Class Honours · GPA: 6.72 / 7
- Excellence Scholarship (2019) · CECC Dickins Scholarship (2023)

SKILLS

AI & ML: LLM evaluation, RAG, agentic LLM systems, AI control (ControlArena, Inspect), LangChain, Ollama
ML libraries: PyTorch, TensorFlow, scikit-learn, Whisper.cpp, librosa
Software: Python, C++, Java, MATLAB, FastAPI, Docker, VirtualBox, Flutter, Figma, LaTeX
Robotics & research: ROS, Gazebo, ArduPilot, STM32, I2C, Altium (PCB design), state estimation, sensor fusion, Lie theory

OTHER EXPERIENCE



Freelance Web Designer
pivotr studio | Sep 2026 – Present

- Clean, fast websites with no ongoing fees, and one free build every month for a non-profit

Private Tutor
Math & Physics | 2019 – Present

- Tutored 60+ students since 2019, Yr 3 to 12, from diverse backgrounds
- Improved student grades from C/B to A average
- Simplified complex concepts, taught time management and self-sufficiency, and focused on instilling lasting subject curiosity

Testing Analyst
BixeLab | Oct 2021 – Nov 2023

- Data analysis on biometric tests to inform client reports
- Automated internal processes and handled risk management
- Delivered 15+ projects, running 3–5 at a time and adapting quickly to new work (small team of 15)

Acoustic Emotion Recognition with ML
ANU R&D Student | 2021

sklearn · TensorFlow · librosa

- Investigated spatial audio (HRTF) features for acoustic emotion recognition with ML (Python, MATLAB, librosa)
- Implemented and adapted KNN, MLP and CNN models (sklearn, TensorFlow), with data visualisation in Seaborn

EXTRA-CURRICULAR

Workstay Volunteer · Workaway | Apr 2026

- Volunteered for 2 weeks helping a family in a small Catalan town build a fence and start construction of their garage

Head of Social · ANU Engineers Assoc. | 2020 – 2021

- Led a team of 3 to run 5 social events for ANU's 3rd largest student body (~$2,000 in sales, up on the previous year)

Community Volunteer · Aus Sri Lankan Buddhist Assoc. | 2010 – Present

- MC at cultural events, and regular volunteer at community events (logistics, photography)

National Premier League Football · Player | 2013 – 2021

- Played at the highest level in Canberra, and team captain for multiple seasons
- Runners-up in 2019 (U18 NPL1) and semi-finalists in 2021 (U23 NPL2)

References available on request.